

Everything as a service

A consumer in San Francisco hunts for a restaurant's address, a small-business woman in Paris checks textile prices in Bangalore, and a financial services executive in London studies global stock market trends. This insatiable worldwide need for real-time information has been fuelling Hewlett Packard's strategy to converge cloud computing and the connectivity of mobile personal information devices. The result of this decade-long effort is the mobility/cloud ecosystem, which promises to deliver personalised experiences through a scalable and secure information infrastructure.

The research agenda to achieve this goal ultimately aims to deliver:

- Infrastructure as a Service — technologies for servers, storage, networking and IT management
- Platform as a Service — technologies for a shared cloud infrastructure that provides enterprise-grade security, scalability and quality of service
- Software as a Service — novel consumer and enterprise services for the cloud

Bringing this vision to life are modern mobile operating systems like Palm webOS, combined with the ability to deliver secure, scalable computing resources economically and on demand. HP already provides millions of internet users with on-demand publications, thousands of webOS mobile phone applications and more. The next step is to build and expand a secure and scalable cloud infrastructure using servers, storage, networks and client devices such as printers, PCs and mobile phones. This convergence of technology and services will be the basis for the next generation of extremely high performing, low-cost connected devices supporting personalised anytime, anywhere access to information and services.

Infrastructure as a Service

Computing infrastructure is undergoing a revolution, characterised by large-scale data centres with millions of users accessing thousands of servers. Such data centres present unique challenges for server design, the ability to scale server configurations and the need to mitigate enormous capital and operating

expenses associated with power and cooling.

As part of our research on exascale data centres, we've developed workloads and metrics to use in building and analysing new system architectures optimised for the cloud. We've optimised our microblade and megaserver architecture for the cloud, and evaluation results show that they could improve energy efficiency by a factor of 4:6 relative to current technology.

Emerging non-volatile memories like memristors combined with advances in photonics and multicore processing offer intriguing opportunities for new system designs, such as nanostores, that could offer significantly better performance and energy efficiency. These improvements in future system architectures will pave the way for applications previously not possible in the cloud, enabling a more sophisticated generation of insights across diverse data sources.



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Platform as a Service

Offering a platform as a service means providing an enterprise-grade cloud computing infrastructure — a service that enforces quality-of-service guarantees over the security, isolation, reliability and performance of the virtualized infrastructures that it generates and manages. Service providers need a platform that can construct multi-tiered, flexible virtual infrastructures providing few restrictions relative to existing physical infrastructures. Cloud services need to be flexible and scalable (though with a limited scope).

To meet service providers' objectives, we took a Cells-as-a-Service approach, introducing a class of virtualized infrastructure services, in which the user declares the desired topology of the virtual machines, virtual block storage and virtual networking. Our research focuses on the requirements for an enterprise-grade cloud computing infrastructure.

Software as a Service

Web 2.0 and the cloud have given rise to a new class of services that cater to an increasingly connected population. HP aims to create models, methods and technologies to harness the flow of collective attention, supporting a mobile society with context-aware and anticipative solutions. The ultimate goal is to build a fluid enterprise that captures collective intelligence for a variety of uses, from predicting the future to allocating resources. Some of our related research projects include ePrint, i-Catcher and Watercooler.

The convergence of cloud computing and connectivity is a reality. Many challenges remain, however, including securing services, data and the infrastructure from attack, ensuring predictable performance as well as ensuring the privacy of personal data. And a generation of scientists and engineers will be focused on creating and delivering these advances. From San Francisco to Paris to Bangalore to London, HP, our customers and partners are working to open the information economy to more people than ever before.

P. Banerjee, C. Bash, R. Friedrich, P. Goldsack, B. Huberman, J. Manley, C. Patel, P. Ranganathan, and A. Veitch, "Everything as a Service: Powering the New Information Economy," IEEE Computer, March 2011, vol. 44, no. 3, pp. 36-43. 